

Solar Power Open Access (Telangana)

Executive Summary

This document provides a comprehensive overview of **solar power Open Access projects in Telangana**, with a specific focus on **third-party Open Access, captive, and group captive business models**.

The purpose is to clearly explain:

- How solar power is generated and connected to the grid
- How electricity flows from a solar plant to industrial consumers
- How commercial settlement, charges, and savings are calculated
- What risks exist (such as curtailment and evacuation constraints) and how they can be managed

The document is intended to serve as a **single reference** for stakeholders evaluating, developing, investing in, or consuming power from Open Access solar projects in Telangana.

Purpose of the document

The purpose of this document is to provide a **structured and practical understanding of solar Open Access projects in Telangana**, covering both **business and technical aspects** in a single reference.

The document is designed to support:

- **Strategic decision-making** (choice of business model, location, voltage level)
- **Commercial evaluation** (savings, charges, risks)
- **Project planning and execution** (evacuation, approvals, compliance)

It is **not a legal or regulatory opinion**, but a practical guide based on prevailing regulations and industry practice in Telangana.

Scope of Coverage

This document covers the **end-to-end lifecycle** of a solar Open Access project in Telangana, including:

- Overview of the Telangana power sector and grid structure
- Definitions and explanations of key electrical, commercial, and regulatory terms
- Detailed explanation of:
 - Third-party Open Access
 - Captive Open Access
 - Group Captive Open Access
- Solar power plant technical architecture, from panels to grid injection
- Voltage levels, evacuation planning, and substation connectivity
- Scheduling, metering, energy accounting, and billing
- Applicable charges, losses, banking, and settlement mechanisms
- Curtailment, congestion, and grid-related risks
- Contracting, compliance, and verification requirements
- A complete end-to-end case study demonstrating how power flows from a plant to multiple consumers

The focus is on **industrial and commercial Open Access**, rather than utility-scale DISCOM power procurement.

Glossary and Key Terminology

This section defines the key technical, commercial, and regulatory terms used throughout this document.

Power and Energy Units

1. kW (kilowatt), MW (megawatt), GW (gigawatt)

These units represent **power**, which is the *instantaneous rate* at which electricity is generated or consumed.

Example: A 2 MW solar plant can deliver **up to 2 MW of power at any given moment**, depending on sunlight.

2. kWh (kilowatt-hour), MWh (megawatt-hour), MU (million units), TWh

These units represent **energy**, which is power measured **over time**.

Energy units are used for **billing, accounting, and settlement**.

3. Relationship between power and energy:

Energy (kWh) = Power (kW) × Time (hours)

Example: If a 2 MW plant operates at full output for 5 hours: $2 \text{ MW} \times 5 \text{ hours} = \mathbf{10 \text{ MWh} = 10,000 \text{ kWh (10,000 units)}}$

1 MU = 1,000,000 kWh

Solar Capacity Terms

1. MWp (Megawatt-peak)

MWp refers to the **rated capacity of solar panels on the DC side**, measured under standard test conditions.

It represents the *maximum potential output* of the panels under ideal sunlight.

MWp is used for:

- EPC design
- Land sizing
- Estimating annual energy generation

2. MWAC (AC Capacity)

MWAC refers to the **maximum real power that can be exported to the grid**, limited by:

- Inverter capacity
- Grid approval

Even if more DC capacity is installed, the plant **cannot inject more than its approved MWAC**.

3. DC/AC Ratio

The DC/AC ratio indicates how much **DC panel capacity** is installed relative to **AC export capacity**.

Formula:

DC/AC Ratio = Installed DC Capacity (MWp) ÷ AC Export Capacity (MWAC)

Example:

If a plant has:

13 MWp of panels

10 MWAC of inverter capacity

DC/AC ratio = $13 \div 10 = 1.3$

What it does in practice:

- Improves generation during morning and evening hours
- Increases annual energy output
- Causes some energy loss at peak sunlight (clipping)
- In Telangana, typical DC/AC ratios for utility-scale solar range from **1.2 to 1.35**.

Electrical System Terms

1. kVA (kilovolt-ampere), MVA (megavolt-ampere)

These units represent **apparent power**, which is the total electrical capacity equipment must handle.

kVA/MVA ratings are used for:

- Inverters
- Transformers
- Switchgear
- Substation bays

They define the **electrical stress capacity**, not just usable power.

2. Power Factor (PF)

Power factor indicates how effectively apparent power (kVA) is converted into usable power (kW).

Formula:

Power Factor = Real Power (kW) ÷ Apparent Power (kVA)

PF ranges from **0 to 1**

A PF closer to 1 means more efficient usage

Example:

An inverter rated at 10 MVA with PF = 0.98 can export:

$10 \times 0.98 = 9.8 \text{ MW}$

Typical PF for solar plants:

- Solar inverters are generally configured between **0.95 and 0.99**
- Most grid approvals in Telangana require **0.98 or above**, with dynamic control capability

3. Reactive Power (Q)

Reactive power is used to **support voltage and magnetic fields** in the electrical system.

- It does not produce usable energy
- It is required for transformers, motors, and transmission lines
- Solar inverters inject or absorb reactive power to help **control voltage**

Poor reactive power control can lead to **voltage rise and curtailment**.

Performance and Planning Metrics

1. CUF (Capacity Utilization Factor)

CUF measures how much energy a plant actually generates compared to its theoretical maximum.

Formula:

$$\text{CUF} = (\text{Actual Energy Generated in a Period}) \div (\text{Installed Capacity} \times \text{Total Hours in that Period})$$

Example (annual):

A 10 MW plant generates 17.5 million units in a year

Maximum possible = $10 \times 8,760 = 87.6$ million units

$\text{CUF} = 17.5 \div 87.6 \approx \mathbf{20\%}$

Typical solar CUF in Telangana:

- Fixed-tilt plants: **18% – 20%**
- Single-axis tracking plants: **20% – 22%**

2. Equivalent Sun Hours

Equivalent sun hours represent the number of hours per day during which solar radiation is equivalent to full sunlight.

In Telangana, typical values range from **4.5 to 5.5 hours per day**, and are commonly used for quick daily energy estimates.

Grid and Network Terminology

1. DISCOM (Distribution Company):

Responsible for delivering electricity to consumer

2. TRANSCO / STU (State Transmission Utility):

Manages transmission networks of 132 kV, 220 kV, and above

Is a Bulk power corridors between regions

3. SLDC (State Load Dispatch Centre):

SLDC is responsible for:

- Scheduling power
- Maintaining grid stability
- Issuing curtailment instructions

4. Substation, Feeder, Bay

Substation: Facility for voltage transformation and switching

Feeder: Line carrying power from a substation

Bay: Dedicated connection point within a substation for a generator or line

Commercial and Regulatory Terms

1. Open Access

Open Access allows a consumer to procure electricity from a generator of their choice using the state grid, instead of buying entirely from the local DISCOM.

2. Wheeling

Wheeling refers to the use of the grid to transport power from the generator to the consumer.

DISCOMs charge wheeling fees for this service.

3. Transmission and Wheeling Losses

Losses occur due to:

- Electrical resistance in lines
- Transformer losses
- Multiple voltage conversions

Typical loss ranges in Telangana (indicative):

- Transmission losses: **2% – 4%**
- Wheeling (distribution) losses: **2% – 5%**

Actual losses depend on:

- Voltage level
- Distance
- Network configuration

4. Cross Subsidy Surcharge (CSS)

CSS is charged to compensate DISCOMs when consumers procure power from outside sources.

Telangana (2025 practice):

CSS varies by:

- Consumer category
- Voltage level

Typical range: ₹1.2 – ₹2.5 per unit

CSS is **not applicable** for captive and group captive consumption when compliance conditions are met.

5. Additional Surcharge (AS)

AS recovers fixed costs incurred by DISCOMs due to long-term power procurement.

Telangana (2025 practice):

- AS is notified periodically by TGERC
- Typical range when applicable: ₹0.5 – ₹1.0 per unit

AS is also **waived for captive and group captive projects**, subject to compliance.

6. Banking

Banking allows renewable energy generated at one time to be **adjusted against consumption at another time**, as per regulatory limits.

It is a **commercial settlement mechanism**, not physical storage.

7. Curtailment

Curtailment occurs when SLDC instructs a generator to reduce output due to grid constraints such as congestion or voltage issues.

1.Stakeholders and Roles

This section describes the key entities involved in a solar Open Access project and clearly defines **who is responsible for what** during development, execution, and operation.

Understanding these roles is critical for:

- Project structuring
- Risk allocation
- Contracting and compliance
- Smooth coordination with grid authorities

1.1 Project Entities

1.1.1 Developer

The **Developer** is the entity that **conceptualizes and drives the project** from idea to operation.

Key responsibilities of the Developer include:

- Identifying suitable land and project location
- Deciding the business model (third-party, captive, or group captive)
- Structuring commercial arrangements with consumers
- Setting up the project SPV
- Coordinating financing, EPC, and O&M
- Managing regulatory approvals and stakeholder relationships

The Developer typically **does not inject power directly** into the grid.

Instead, it operates through a dedicated project company (SPV).

1.1.2 Special Purpose Vehicle (SPV)

The **SPV** is a **separate legal entity** created specifically for one solar project.

Key characteristics of the SPV:

- Owns the solar plant assets
- Holds grid connectivity and Open Access approvals
- Signs Power Purchase Agreements (PPAs) with consumers
- Signs EPC and O&M contracts

- Is responsible for regulatory compliance and reporting

In captive and group captive projects:

- Consumers hold **equity shares directly in the SPV**
- Compliance with **26% ownership and 51% consumption** is evaluated at the SPV level

The SPV structure ensures:

- Clear ownership
- Ring-fenced project risk
- Ease of financing and compliance

1.1.3 How are Developer and SPV different?

Developer and **SPV** are related, but they are **not the same entity**.

- The **Developer** is the **company or group** that:
 - Identifies the opportunity
 - Finds land
 - Structures the business model
 - Signs PPAs with consumers
 - Arranges financing
 - Oversees EPC and operations
- The **SPV (Special Purpose Vehicle)** is a **separate legal company** created **only for one specific project**.

Think of it like this: The **Developer is the parent / promoter**, the **SPV is the project company**.

Why an SPV is used

- Keeps **project risks isolated**
- Allows **project-level financing**
- Makes ownership and compliance (especially captive / group captive) clean and auditable
- Regulators, lenders, and consumers contract with the **SPV**, not the developer

Example:

- Developer: ABC Renewables Pvt Ltd

- SPV: ABC Solar Siddipet 20 MW Pvt Ltd

1.1.4 EPC Contractor (Engineering, Procurement, and Construction)

The EPC contractor is responsible for **designing and building the solar plant**.

Typical EPC scope includes:

- Detailed engineering and layout
- Procurement of panels, inverters, transformers, and switchgear
- Construction, installation, and commissioning
- Performance guarantees during construction

The EPC contractor's role usually ends after **successful commissioning**, except for defect liability obligations.

1.1.5 O&M Contractor (Operations and Maintenance)

The O&M contractor is responsible for **day-to-day operation** of the plant after commissioning.

Key responsibilities include:

- Plant monitoring and performance optimization
- Preventive and corrective maintenance
- Grid coordination during outages or curtailment
- Ensuring availability and generation targets

In many cases, the Developer or its affiliate acts as the O&M contractor.

1.2 Grid and System Entities

1.2.1 DISCOM (Distribution Company)

The DISCOM:

- Operates the **distribution network** (typically 11 kV and 33 kV)
- Applies wheeling charges and losses
- Facilitates drawal of power by consumers
- Issues Open Access approvals within its jurisdiction

The DISCOM does **not own the generated power**, but enables its movement and accounting.

1.2.2 TRANSCO / STU (State Transmission Utility)

The transmission utility:

- Operates **high-voltage transmission lines** (132 kV and above)
- Provides evacuation infrastructure for large projects
- Maintains grid reliability across regions

Projects connected at 132 kV or higher interact primarily with TRANSCO.

1.2.3 SLDC (State Load Dispatch Centre)

The SLDC is the **grid operator** and acts as the system coordinator.

Its responsibilities include:

- Scheduling generation and consumption
- Monitoring grid stability
- Issuing curtailment or back-down instructions
- Energy accounting and reporting

All Open Access projects operate under SLDC's operational control.

1.3 Roles and Responsibilities (RACI Framework)

This subsection clarifies **who is responsible, accountable, consulted, and informed** across the project lifecycle.

1.3.1 Development and Approvals

- Developer: Project planning, land acquisition, model selection
- SPV: Filing applications and holding approvals
- DISCOM / TRANSCO: Connectivity and evacuation approvals
- SLDC: Operational feasibility review

1.3.2 Construction and Commissioning

- EPC: Design, construction, testing, commissioning
- SPV: Contract management and payment
- DISCOM / TRANSCO: Grid synchronization and approvals
- SLDC: Trial run coordination

1.3.3 Operations, Scheduling, and Settlement

- O&M: Plant operation and availability
- SPV: Scheduling submissions and compliance
- SLDC: Scheduling approval and curtailment decisions
- DISCOM: Metering, billing, and settlement

1.3.4 Compliance and Reporting

- SPV: Captive compliance, energy accounting, reporting
- Consumers: Consumption tracking (especially for captive / group captive)
- SLDC / DISCOM: Verification and audits

Failure in compliance can lead to:

- Loss of captive status
- Levy of CSS and AS
- Financial impact on consumers

2. Telangana Power Sector Overview

2.1 Telangana Electricity Ecosystem

Telangana's electricity system operates as an **integrated state grid**, where power generated from multiple sources flows through a common transmission and distribution network before reaching consumers.

Key characteristics of the Telangana power ecosystem include:

- Centralized grid operation under the State Load Dispatch Centre (SLDC)
- Integrated operation of generation, transmission, and distribution
- Priority integration of renewable energy sources such as solar and wind
- Separation of **network ownership** (DISCOM vs TRANSCO) from **power flow**

Electricity injected at any point in the grid becomes part of the common pool and is **not physically traced to a specific consumer**. Instead, consumption is settled through **commercial accounting mechanisms**.

2.2 Roles of DISCOM, TRANSCO, and SLDC

The Telangana power sector operates through three distinct but interconnected entities:

DISCOM (Distribution Company)

- Operates distribution networks (typically 11 kV and 33 kV)
- Supplies power to end consumers
- Manages metering, billing, wheeling, and losses

TRANSCO (Transmission Utility / STU)

- Operates high-voltage networks (132 kV and above)
- Enables bulk transfer of power across regions
- Provides evacuation corridors for large generators

SLDC (State Load Dispatch Centre)

- Acts as the system operator
- Schedules generation and consumption
- Ensures grid stability
- Issues curtailment or back-down instructions when required

2.3 Voltage-Level Ownership and Grid Connectivity

Voltage levels determine **which utility owns and operates the network**, not how power ultimately reaches consumers.

- **Up to 33 kV** - Network owned and operated by DISCOM
- **132 kV and above** - Network owned and operated by TRANSCO

However, selecting a connection voltage **does not isolate the power flow** to that voltage level alone.

For example:

- A generator connected at 33 kV may still have its power flow through 132 kV or 220 kV corridors to reach distant consumers
- A generator connected at 132 kV will still rely on DISCOM networks for last-mile delivery to factories

The grid functions as a **single synchronized system**, regardless of the connection voltage.

2.4 How Power Reaches Consumers in Practice

In practical terms:

- Power is injected at the approved interconnection point
- It flows through the integrated grid based on electrical laws, not contracts

- Consumers draw power from their local DISCOM network
- Energy delivery is reconciled through:
 - Injection meters
 - Drawal meters
 - Energy accounting by SLDC and DISCOM

Thus, **all consumers receive power through DISCOM systems**, even when generation is connected to the TRANSCO network.

2.5 Distance, Network Usage, and System Impact

Distance between generator and consumer:

- Does **not usually change fixed charges**, which are regulated
- Does **affect technical losses**, as power flows through:
 - Multiple substations
 - Multiple voltage transformations
 - Longer line lengths

Lower voltage over long distances results in higher losses, while higher voltage transmission reduces losses but increases infrastructure cost.

Detailed discussion on:

- Loss percentages
- Charge calculations
- Distance-related commercial impact

3. DISCOM Structure in Telangana

Telangana's electricity distribution system is divided into **two Distribution Companies (DISCOMs)**. These DISCOMs are responsible for operating the **distribution network**, supplying power to consumers, and facilitating **Open Access drawal and wheeling** within their respective jurisdictions.

Understanding DISCOM boundaries is critical for Open Access projects because **charges, losses, approvals, billing, and coordination** are all influenced by the DISCOM under which the generator and the consumer are located.

3.1 Southern DISCOM – TGSPDCL

Telangana State Southern Power Distribution Company Limited (TGSPDCL) serves the **southern part of Telangana**, including Hyderabad and its surrounding industrial regions.

3.1.1 Regions and Districts Covered

TGSPDCL covers the following districts and regions:

- Hyderabad
- Rangareddy
- Medchal–Malkajgiri
- Sangareddy
- Vikarabad
- Mahabubnagar
- Narayanpet
- Jogulamba Gadwal
- Wanaparthy
- Nagarkurnool
- Nalgonda
- Yadadri Bhuvanagiri
- Suryapet

These regions include a high concentration of **commercial, IT, pharmaceutical, manufacturing, and service-sector consumers**.

3.1.2 Load Profile and Industrial Concentration

Key characteristics of TGSPDCL include:

- **High and relatively stable electricity demand**
- Significant **daytime and round-the-clock industrial load**
- Large number of **Open Access consumers and drawal points**
- Strong presence of:
 - Pharma manufacturing
 - IT parks and data centers
 - Large industrial estates

For Open Access projects, TGSPDCL areas are typically **drawal-heavy**, meaning:

- Power consumption is closer to generation hours
- Banking dependency may be lower

- Revenue realization tends to be more stable

3.2 Northern DISCOM – TGNPDCL

Telangana State Northern Power Distribution Company Limited (TGNPDCL) serves the **northern and central parts of Telangana**.

3.2.1 Regions and Districts Covered

TGNPDCL covers the following districts and regions:

- Adilabad
- Komaram Bheem (Asifabad)
- Mancherial
- Nirmal
- Nizamabad
- Kamareddy
- Karimnagar
- Jagtial
- Peddapalli
- Rajanna Sircilla
- Siddipet
- Hanumakonda
- Warangal
- Jangaon
- Mahabubabad
- Mulugu
- Khammam
- Bhadrachalam

3.2.2 Solar Project Concentration and Land Availability

Key characteristics of TGNPDCL include:

- **Large availability of contiguous land parcels**
- Strong **solar resource availability**

- Lower population density compared to southern Telangana
- Presence of multiple:
 - Utility-scale solar plants
 - Captive and group captive solar projects

As a result, most **solar generation (injection points)** are located in TGNPDCL areas, even when consumers are located in TGSPDCL.

This creates a common Open Access pattern:

- **Generation in TGNPDCL**
- **Consumption in TGSPDCL**
- Power flowing through the common state grid

3.3 Why DISCOM Boundaries Matter for Open Access

DISCOM boundaries do **not restrict physical power flow**, but they **directly affect commercial settlement and coordination**.

Key implications include:

- **Billing and Settlement**
 - Injection is accounted at the generator's DISCOM
 - Drawal is accounted at the consumer's DISCOM
 - Energy is settled through SLDC accounting mechanisms
- **Charges and Losses**
 - Wheeling charges and losses may differ by DISCOM
 - Applicable tariffs are based on the DISCOM orders in force
- **Operational Coordination**
 - Both DISCOMs coordinate with SLDC for scheduling and settlement
 - Curtailment instructions may be influenced by local network constraints
- **Project Structuring**
 - Developers must assess:
 - Which DISCOM the plant falls under
 - Which DISCOM the consumers fall under
 - Impact on losses, banking, and billing complexity

In summary, while electricity physically flows through the **integrated state grid**, Open Access projects are **commercially and administratively governed by DISCOM jurisdictions**, making DISCOM mapping a critical early-stage planning activity.

4. Open Access Business Models

This section explains the three Open Access business models permitted under Indian and Telangana electricity regulations, and how each model differs in terms of ownership, power flow, charges, compliance, and risk.

4.1 Third-Party Open Access

In a third-party Open Access model, a solar power developer sets up a generating plant and sells electricity directly to one or more consumers under a Power Purchase Agreement (PPA). The consumers do **not hold any ownership stake** in the generating project.

Key characteristics:

- Generator and consumer are independent entities
- Power is sold purely under a commercial PPA
- Consumers continue to remain retail consumers of the DISCOM for backup and balancing power

Commercial implications:

- Transmission charges apply
- Wheeling charges apply
- Transmission and wheeling losses apply
- Cross Subsidy Surcharge (CSS) applies
- Additional Surcharge (AS) applies, unless waived by regulatory order

Because CSS and AS are applicable, the net landed cost of power is typically higher than captive models. As a result, third-party Open Access is generally preferred only when:

- Captive compliance is not feasible
- Consumers want zero equity participation
- Regulatory surcharges are temporarily low or waived

4.2 Captive Open Access (Single Consumer)

Captive Open Access allows a **single industrial or commercial consumer** to jointly own a power plant and consume electricity from it.

4.2.1 Ownership Requirement

- The captive consumer must hold **at least 26 percent equity** in the generating project
- The equity must carry **voting rights**

4.2.2 Consumption Requirement

- The captive consumer must consume **at least 51 percent of the total electricity generated**, measured on an **annual basis**

Failure to meet either condition can result in the project losing captive status.

4.2.3 Commercial Benefits

When captive conditions are met:

- Cross Subsidy Surcharge (CSS) is **fully waived**
- Additional Surcharge (AS) is **fully waived**
- Only transmission charges, wheeling charges, and losses apply

This makes captive Open Access one of the **most cost-effective structures** for large power-consuming industries.

Typical users:

- Large manufacturing units
- Continuous process industries
- Data centers
- Cement, steel, pharma, and chemicals

4.3 Group Captive Open Access (Multiple Consumers)

Group Captive Open Access extends the captive concept to **multiple consumers jointly**.

4.3.1 Ownership and Consumption Criteria

- One or more consumers together must hold **at least 26 percent equity** in the project
- The group of captive consumers together must consume **at least 51 percent of the total annual generation**

Each consumer:

- Holds a defined equity share
- Has a defined energy allocation

4.3.2 Allocation and Settlement Across Members

- Energy generated is allocated among captive members based on agreed shares
- Allocation is reflected in SLDC energy accounting statements
- Each consumer is billed separately by the DISCOM for net drawal after Open Access adjustment

4.3.3 Benefits and Operational Complexity

Benefits:

- CSS and AS are waived for captive consumption
- Medium and smaller industries can access captive benefits
- Risk and investment are shared

Complexity:

- Annual compliance monitoring is critical
- Shareholding and energy consumption must be tracked continuously
- Any shortfall in the 51 percent rule can lead to surcharge exposure

Group captive projects require **strong governance and documentation** to remain compliant.

4.4 Comparison of Open Access Models

4.4.1 Commercial Comparison

- Third-party Open Access:
 - CSS applicable
 - AS applicable (unless waived)
 - Lowest compliance burden
 - Higher landed power cost
- Captive Open Access:
 - CSS waived
 - AS waived
 - Medium compliance burden

- Lowest landed power cost
- Group Captive Open Access:
 - CSS waived
 - AS waived
 - Highest compliance and coordination effort
 - Attractive for mid-sized consumers

4.4.2 Risk and Compliance Comparison

- Third-party:
 - Regulatory surcharge risk
 - Lower compliance risk
- Captive:
 - Risk of losing captive status if ownership or consumption thresholds are not met
- Group captive:
 - Highest compliance risk
 - Requires continuous monitoring of equity and energy offtake

5. Core Electrical Concepts for Open Access

This section explains the core electrical concepts that are essential to understand how solar Open Access projects operate on the grid. These concepts directly influence plant sizing, grid approvals, losses, billing, and curtailment risk.

5.1 Power versus Energy in Practical Terms

Power refers to the **rate** at which electricity is generated or consumed at a given moment.

It answers the question: *“How much electricity is flowing right now?”*

- Measured in kW or MW
- Used to describe:
 - Plant capacity (e.g., 2 MW plant)
 - Consumer demand (e.g., 5 MW industrial load)
 - Grid limits and evacuation capacity

Energy refers to the **total amount** of electricity generated or consumed over time.

It answers the question: *“How much electricity was used or generated over a period?”*

- Measured in kWh, MWh, or MU
- Used for:
 - Billing
 - Energy settlement
 - Banking calculations

Relationship between power and energy:

Energy (kWh) = Power (kW) × Time (hours)

Example:

If a factory draws 1 MW continuously for 10 hours, it consumes **10 MWh (10,000 units)**.

5.2 DC Generation and AC Grid Connection

Solar panels generate **Direct Current (DC)** electricity.

However, the grid and industrial consumers operate on **Alternating Current (AC)**.

This creates a two-stage system:

- DC power is generated at the panel level
- DC power is converted to AC before being injected into the grid

Key implications for Open Access:

- Panel capacity is expressed in **MWp (DC)**
- Grid injection is limited by **MWAC (AC)**
- Grid approvals are always based on **AC capacity**

This DC-to-AC distinction is fundamental to understanding:

- Inverter sizing
- Clipping
- Export limits approved by DISCOM or TRANSCO

5.3 Inverters, kVA Rating, and Apparent Power

Inverters convert DC power from solar panels into AC power suitable for the grid.

Inverters are rated in **kVA or MVA**, not in kW or MW.

This is because electrical equipment must be designed to handle **total electrical load**, not just usable power.

- **kVA (apparent power)**: total electrical capacity handled by equipment
- **kW (real power)**: actual usable power delivered

The relationship is defined by **Power Factor (PF)**.

Example:

An inverter rated at 10 MVA operating at a PF of 0.98 can deliver:

$$10 \times 0.98 = \mathbf{9.8 \text{ MW of real power}}$$

For Open Access projects:

- Inverter MVA rating must align with grid approval
- PF requirements are specified by DISCOM/SLDC
- Incorrect sizing can lead to penalties or curtailment

5.4 Power Factor and Grid Requirements

Power Factor (PF) indicates how efficiently electrical capacity is being used.

- PF ranges from 0 to 1
- A PF close to 1 indicates efficient operation

Formula:

$$\text{Power Factor} = \text{Real Power (kW)} \div \text{Apparent Power (kVA)}$$

For solar plants:

- Typical operating PF is **0.95 to 1.0**
- Grid operators may instruct plants to:
 - Operate at a lower PF
 - Absorb or inject reactive power
- This is done to control voltage and maintain grid stability

Operating outside approved PF limits can result in:

- Grid non-compliance
- Reduced export capacity
- Curtailment instructions

5.5 Reactive Power and Voltage Control

Reactive power is required to maintain **voltage levels** in the electrical system.

- It does not produce usable energy
- It supports:
 - Magnetic fields in transformers
 - Voltage stability on transmission lines

In solar Open Access projects:

- Inverters are required to provide **reactive power support**
- SLDC may instruct plants to:
 - Absorb reactive power during high-voltage conditions
 - Inject reactive power during low-voltage conditions

Poor reactive power management can cause:

- Voltage rise issues
- Grid instability
- Forced reduction in active power output (curtailment)

Reactive power control is therefore a **grid support obligation**, not an optional feature.

6. Solar Plant Technical Architecture

This section explains the technical components of a solar power plant, starting from electricity generation at the panel level and ending at the point where power is ready for evacuation to the grid.

The focus is on how each component supports grid injection, metering, and compliance in an Open Access context.

6.1 Solar PV Generation (DC Side)

Solar panels generate electricity in direct current (DC) form when sunlight falls on the photovoltaic cells.

Each panel produces:

- Low-voltage DC power
- Output varies with sunlight intensity, temperature, and shading

Panels are installed in large numbers and must be electrically combined in a safe and organized manner before power can be sent to the inverter.

6.1.1 Strings and Combiner Boxes

Strings

- A string is a group of solar panels connected in series
- Series connection increases voltage to a usable level
- Typical string voltage ranges from 800 V to 1,500 V DC

Combiner Boxes

- Multiple strings are connected in parallel inside a DC combiner box
- The combiner box:
 - Collects DC power from many strings
 - Provides protection using fuses and surge protection devices
 - Sends a single, higher-current DC output to the inverter

Simple analogy:

Many small water pipes (strings) are joined into a larger pipe (combiner box) so water can be transported efficiently to the pump (inverter).

6.2 Inverters and Control Philosophy

The inverter is one of the most critical components in a solar plant.

Its main functions are:

- Convert DC power to AC power
- Match grid voltage and frequency
- Control power factor and reactive power
- Respond to SLDC grid instructions

In Open Access projects, inverter configuration must strictly follow grid codes and SLDC requirements.

6.2.1 Power Factor (PF) Setpoint and Reactive Capability

Power Factor Setpoint

- The PF setpoint is a control value programmed into the inverter
- It defines how much real power (kW) versus reactive power (kVAr) the inverter supplies
- Typical solar inverter PF range: 0.95 to 1.0

Why this matters

- SLDC may instruct plants to operate at a specific PF to control grid voltage

- Poor PF control can lead to:
 - Voltage rise
 - Grid instability
 - Curtailment instructions

Modern inverters are capable of dynamic PF adjustment without reducing real power output within limits.

6.3 Transformers and Voltage Step-Up System

The AC power from inverters is generated at low or medium voltage, typically:

- 415 V
- 690 V
- 3.3 kV

Before power can be evacuated, voltage must be increased using step-up transformers.

Transformer functions

- Increase voltage to match grid connection level (11 kV, 33 kV, 132 kV, etc.)
- Reduce current for efficient transmission
- Provide electrical isolation and protection

Transformer selection affects:

- Losses
- Reliability
- Long-term operational cost

6.4 Switchyard and Protection System

The switchyard is the interface between the solar plant and the grid.

It ensures safe connection, disconnection, protection, and metering.

6.4.1 Breakers and Isolators

Circuit Breakers

- Automatically disconnect the plant during:
 - Short circuits
 - Over-voltage

- Frequency deviations
- Protect both plant equipment and the grid

Isolators

- Provide visible electrical separation
- Used for maintenance and safety
- Cannot interrupt load (only operated when breaker is open)

6.4.2 CT/PT and Protection Relays

CT (Current Transformer)

- Measures current flowing into the grid

PT (Potential Transformer)

- Measures voltage at the connection point

Protection Relays

- Continuously monitor electrical parameters
- Detect abnormal conditions
- Trip the breaker when limits are exceeded

These devices ensure compliance with grid protection standards.

6.4.3 Metering Interface

Meters installed at the grid interface:

- Measure exported energy
- Record data in 15-minute time blocks
- Provide input for:
 - Scheduling
 - Billing
 - Energy accounting

Meters used for Open Access are typically ABT / SEM compliant and sealed by the utility.

6.5 DC/AC Ratio and Clipping

DC/AC Ratio

- Ratio of total panel capacity (MWp) to inverter AC capacity (MWAC)
- Typical values in Telangana: 1.1 to 1.4

Why DC is oversized

- Solar panels rarely operate at peak output all day
- Extra DC capacity improves:
 - Morning generation
 - Evening generation
 - Annual energy yield

Clipping

- Occurs when DC generation exceeds inverter AC capacity
- Excess energy is not exported and is “clipped”
- Happens mainly during peak sunlight hours

Commercial impact

- Some clipping is acceptable and planned
- Excessive clipping combined with grid curtailment can reduce revenue
- DC/AC ratio must be optimized based on:
 - CUF
 - Evacuation capacity
 - Curtailment risk

7. Voltage Levels and Grid Ownership

Electricity generated from a solar power plant must be evacuated through the grid at an approved voltage level.

The choice of voltage determines:

- Which utility (DISCOM or TRANSCO) the project interfaces with
- How much power can be carried
- Loss levels

- Curtailment risk
- Cost of evacuation infrastructure

7.1 Voltage Hierarchy (LT, 11 kV, 33 kV, 132 kV, 220 kV)

Electric power flows through the grid in a stepped-up and stepped-down manner to reduce losses and manage capacity.

- **LT (415 V / 0.415 kV)**

Used inside consumer premises and small rooftop systems

- **11 kV**

Local distribution voltage, suitable only for very small generators

- **33 kV**

Sub-transmission voltage, commonly used for small and medium solar plants

- **132 kV**

Primary transmission voltage, preferred for large solar projects

- **220 kV and above**

Bulk transmission for large renewable parks and interstate flow

Higher voltage allows the same power to be transmitted with **lower current**, which reduces losses and thermal stress on lines.

7.2 DISCOM versus TRANSCO Responsibility by Voltage

Grid ownership and approvals depend on the voltage level at which the plant connects.

- **DISCOM jurisdiction**

- LT, 11 kV, and typically **33 kV**
- Distribution substations and feeders
- Wheeling charges and wheeling losses apply

- **TRANSCO (STU) jurisdiction**

- **132 kV and above**

- Transmission substations, bays, and corridors
- Transmission charges and transmission losses apply

Important clarification:

Even if a project connects at **132 kV or above (TRANSCO)**, the power is still ultimately **delivered to consumers through DISCOM networks**.

Power does **not bypass the DISCOM**; only the injection point shifts to the transmission system.

7.3 Voltage Selection Considerations

The selection of voltage is a **strategic project decision**, not just a technical one.

7.3.1 Cost and Technical Losses

- Lower voltage:
 - Lower upfront infrastructure cost
 - Higher current
 - Higher technical losses
- Higher voltage:
 - Higher upfront cost (lines, bays, protection)
 - Lower current
 - Lower losses over distance

Distance matters:

- Longer distances increase losses
- Losses are applied **as percentages**, not per-km billing
- Higher voltage reduces loss impact over distance

Losses occur due to:

- Electrical resistance in conductors
- Transformer losses
- Reactive power flow

(Actual loss percentages are defined in tariff orders and vary by voltage level.)

7.3.2 Curtailment and Congestion Risk

Curtailment risk is closely linked to voltage choice:

- **33 kV networks**
 - Designed primarily for load, not generation
 - Limited headroom for reverse power flow
 - Higher voltage rise issues at noon
 - Higher curtailment probability in solar-dense areas
- **132 kV and above**
 - Designed for bulk power flow
 - Better absorption of generation
 - Lower congestion risk
 - Preferred for large Open Access and group captive projects

Choosing a higher voltage does **not eliminate curtailment**, but it **significantly reduces risk**.

7.4 Practical Power Carrying Capacity by Voltage Level

The table below shows **industry-standard indicative capacities** per voltage level.

Actual capacity depends on conductor size, line configuration, ambient conditions, and grid design.

Voltage Level	Typical Practical Capacity
11 kV	5 - 8 MW
33 kV	20 - 40 MW
132 kV	100 - 150 MW
220 kV	300 MW and above

Implications for project planning:

- A 2–10 MW project may use 33 kV if the substation has headroom
- Projects above ~20 MW are better suited for 132 kV
- Solar parks and aggregation hubs prefer 132 kV or 220 kV pooling substations

8. Evacuation and Grid Interconnection

Evacuation and grid interconnection define how electricity generated at a solar power plant is physically delivered into the state electricity network. This is one of the most critical aspects of an Open Access project, as evacuation constraints directly affect project viability, curtailment risk, and timelines.

8.1 Meaning and Importance of Evacuation

Evacuation refers to the **physical infrastructure and approvals required to move power from the solar plant to the nearest grid interconnection point**, typically a substation.

Evacuation includes:

- Electrical equipment within the plant
- Lines connecting the plant to the grid
- Interconnection facilities at the substation

A project may have sufficient generation capacity, but without reliable evacuation, the power **cannot be injected into the grid**, resulting in stranded assets or frequent curtailment.

8.2 Developer Scope and Responsibilities

The developer (through the SPV) is responsible for **planning, building, and coordinating evacuation facilities** up to the designated interconnection point.

8.2.1 Plant-Side Infrastructure

This includes all equipment required to convert solar energy into grid-ready power:

- **Inverters and inverter transformers**

Convert DC power generated by solar panels into AC power at a medium voltage level.

They also control power factor and reactive power to meet grid requirements.

- **Main step-up transformers**

Increase voltage from inverter level (typically 800 V–33 kV) to the evacuation voltage (33 kV or 132 kV).

Higher voltage reduces current and minimizes transmission losses.

- **Internal switchgear and protection systems**

Include circuit breakers, isolators, relays, and protection logic that detect faults and isolate equipment to prevent damage.

- **Control room, SCADA, and communication systems**

Monitor generation, voltage, frequency, breaker status, and grid instructions.

Enable remote control and real-time data sharing with SLDC.

8.2.2 Evacuation Line

The evacuation line is the **dedicated power line connecting the solar plant to the nearest grid substation**.

- Can be:
 - 11 kV
 - 33 kV
 - 132 kV or above
- Typically built and paid for by the developer

Key aspects:

- May be overhead or underground
- Designed for approved voltage and capacity
- Requires **Right of Way (RoW) permission**

Right of Way (RoW) permission is the legal approval to route power lines across:

- Private land
- Government land
- Roads, canals, or forest areas

Delays in RoW are one of the **most common causes of project schedule overruns**.

8.2.3 Substation Interconnection

The method of interconnection depends on **voltage level**.

8.3 Substation Bay Configuration and Constraints

Each substation has:

- Limited physical space
- Limited electrical capacity
- Limited fault-handling capability

Constraints arise due to:

- No spare bays available
- Transformer loading limits
- Short-circuit level limitations
- High solar penetration in the area

In solar-heavy regions of Telangana (north and west districts), bay availability is a frequent constraint.

8.3.1 Substation Interconnection: Bay Requirement by Voltage

A **substation bay** is a dedicated and protected connection point within a substation that allows a generator to inject power safely into the grid.

Bay requirement by voltage level

Voltage Level	Utility	Bay Requirement	Typical complexity
11 kV	DISCOM	Feeder bay	Low
33 kV	DISCOM	Line/feeder bay	Medium
132 kV 220 kV+	TRANSCO TRANSCO	Dedicated bay Dedicated bay	High Very High

At **33 kV**, generators are typically connected via:

- Dedicated feeder
- Breaker panel within DISCOM substation

A full transmission bay is **not mandatory** at this level, though utilities may impose specific configurations.

At **132 kV and above**, a **dedicated bay is mandatory** because the connection is part of the transmission network operated by TRANSCO.

8.3.2 Substation Bay – Components and Functions

A transmission bay includes the following key components:

- **Circuit Breaker**

Automatically disconnects the plant during faults or abnormal conditions.

- **Isolators**

Provide visible separation for maintenance and safety.

- **Current Transformers (CTs)**

Measure current for protection and energy metering.

- **Potential Transformers (PTs)**

Measure voltage for protection, control, and metering.

- **Protection Panels**

Analyze signals from CTs/PTs and initiate tripping during faults.

- **Metering Interface**

Houses energy meters used for billing and settlement.

These components ensure **safe, measurable, and controllable power injection** into the grid.

8.4 Technical Checks Prior to Grid Injection

Before commissioning, the utility and SLDC conduct technical tests to ensure grid safety:

- **Protection coordination test**

Confirms that plant protection operates in coordination with grid protection.

- **Relay testing**

Verifies correct fault detection and tripping logic.

- **CT/PT polarity and ratio test**

- Verifies accurate measurement of current and voltage

- **Synchronization test**

Ensures voltage, frequency, and phase match the grid before connection.

- **SCADA and telemetry test**

Confirms real-time data flow to SLDC.

- **Meter accuracy test**

Ensures energy measurement meets regulatory standards.

These tests prevent:

- Equipment damage
- Grid instability
- Incorrect energy accounting

8.5 Approvals and Commissioning Process

Typical approvals include:

- Grid connectivity approval
- Bay allocation approval
- Evacuation line approval
- Installation and testing
- Trial run permission
- Commercial operation date (COD)

Approvals are issued jointly by:

- DISCOM (for 11 kV / 33 kV)
- TRANSCO (for 132 kV and above)
- SLDC (operational clearance)

8.6 Evacuation Bottlenecks and Mitigation Strategies

Common bottlenecks include:

- **Substation capacity constraints**
Existing substations may not have spare capacity.
- **Transmission corridor congestion**
High solar concentration areas may experience backfeeding issues.
- **Right of Way delays**
Land access issues can halt line construction.
- **Bay availability constraints**
Bay construction may require substation expansion.

Mitigation strategies include:

- Selecting substations with planned augmentation
- Opting for higher voltage evacuation where feasible

- Early RoW planning and stakeholder engagement
- Phased commissioning aligned with grid upgrades
- Designing flexible evacuation routes
- Conservative assumptions on curtailment in PPAs

9. Scheduling, Metering, and Energy Accounting

This section explains how electricity generated by a solar Open Access plant is **scheduled, measured, allocated, and commercially settled** between the generator and consuming industries.

Although electricity physically flows through a **common interconnected grid**, Open Access transactions are managed entirely through **scheduling, metering, and accounting mechanisms**, coordinated by the **State Load Dispatch Centre (SLDC)** and the respective **DISCOMs**.

9.1 Scheduling and Time-Block Concept

In the Indian power system, electricity generation and consumption are scheduled and accounted for in **15-minute time blocks**, commonly referred to as **time blocks**.

- Each day consists of **96 time blocks** (24 hours × 4 blocks per hour).
- All Open Access energy accounting is done on a **time-block basis**, not as a daily lump sum.

For Open Access solar plants:

- The generator submits an **expected generation schedule** to the SLDC, typically **one day in advance**.
- The schedule follows the expected solar generation pattern:
 - Low generation in the morning
 - Peak generation around midday
 - Zero generation during night hours

Scheduling serves multiple purposes:

- Helps SLDC maintain **real-time balance** between generation and demand
- Enables **transparent and auditable energy accounting**
- Forms the basis for **commercial settlement and deviation analysis**

Solar and wind generators are generally classified as “**must-run**” **resources**, meaning:

- They are **not curtailed for economic reasons**
- However, they **may still be curtailed** due to:

- Grid safety concerns
- Network congestion
- Voltage or frequency constraints

9.2 Injection and Drawal Metering

Electricity under Open Access is measured at **two separate locations**, each serving a distinct purpose.

Injection Point (Generation Side)

- Located at the point where the solar plant connects to the grid (substation or feeder).
- Measures:
 - Energy injected into the grid (kWh)
 - Electrical parameters such as voltage, current, and frequency
- Typically equipped with:
 - **Special Energy Meters (SEM)** or
 - **ABT-compliant meters**
- Meter data forms the **official record of generation** for settlement.

Drawal Point (Consumer Side)

- Located at the consumer's point of supply (factory, industrial unit, data center, etc.).
- Measures:
 - Energy drawn from the grid by the consumer
- Used for:
 - Allocation of Open Access energy
 - Billing and settlement

All Open Access meters:

- Are **time-synchronized**
- Are **sealed and approved** by DISCOM and SLDC
- Are read at regular intervals, typically **monthly**, for settlement purposes

9.3 Energy Allocation to Multiple Consumers

In **captive and group captive Open Access projects**, energy generated by a single solar plant is **commercially allocated** across one or more consuming entities.

Key principles of allocation:

- Allocation is **commercial, not physical**
- Electricity is **not traced wire-by-wire** from plant to factory
- Allocation is based on:
 - Metered generation at the injection point
 - Metered consumption at drawal points
 - Agreed allocation ratios
 - Compliance with captive or group captive rules

Captive Open Access

- A single consumer must consume **at least 51 percent** of the plant's total annual generation.
- Allocation is straightforward, as there is only one captive user.

Group Captive Open Access

- Captive users must collectively:
 - Hold **minimum 26 percent equity** in the project
 - Consume **minimum 51 percent** of total annual generation
- Energy is allocated consumer-wise based on:
 - Agreed allocation percentages
 - Actual drawal recorded at each consumer meter

Third-Party Open Access (Non-Captive)

- Energy generated by the plant is allocated to consumers strictly as per:
 - Power Purchase Agreement (PPA)
 - Actual drawal recorded at consumer meters
- There are **no ownership or minimum consumption requirements**
- Consumers pay applicable:

- Transmission and wheeling charges
- Losses
- Cross Subsidy Surcharge (CSS)
- Additional Surcharge (if applicable)

If a consumer draws **less energy than their allocated share** during a settlement period:

- Excess energy may be:
 - **Banked**, if permitted by regulation
 - **Lapsed**
 - **Settled as per PPA and applicable rules**

9.4 Settlement Process and Reconciliation

Settlement is the process of converting **metered energy data** into **commercial bills, credits, and adjustments**.

A typical monthly settlement cycle includes:

1. Meter data collection from:
 - Injection point meters
 - Drawal point meters
2. Validation of data by DISCOM and SLDC
3. Application of:
 - Transmission losses
 - Wheeling losses
4. Banking adjustments, if applicable
5. Allocation of net energy to each consumer
6. Calculation of charges, credits, and adjustments
7. Issuance of final settlement statements

Key characteristics of the settlement process:

- Entirely **data-driven**

- Based on **approved meters and SLDC records**
- Transparent and auditable

Any disputes related to settlement are resolved using:

- Meter data
- SLDC schedules
- Approved regulatory procedures

Important Clarification: Physical Flow vs Commercial Settlement

- **Physical power flow** follows electrical laws:
 - Power flows based on network impedance and grid conditions
 - It is not controllable on a consumer-specific basis
- **Commercial settlement** is independent of physical flow and is based on:
 - Meter readings
 - Approved schedules
 - Allocation rules
 - Regulatory provisions

Consumers pay for **net allocated energy**, not for the physical electrons flowing from a specific plant.

10. Commercial Charges and Billing

This section explains the various **charges, losses, and adjustments** applied to solar Open Access power in Telangana, and how they together determine the **final landed cost of electricity** for consuming industries.

While solar generation itself may have a fixed tariff under a PPA, the **actual cost savings** depend on how grid charges, losses, and regulatory levies are applied.

10.1 Transmission and Wheeling Charges

Transmission Charges

Transmission charges are levied for using the **high-voltage transmission network** operated by the State Transmission Utility (TRANSCO).

These charges apply when power flows through:

- 132 kV and above networks
- State or regional transmission corridors

Transmission charges are:

- Typically levied on a **₹/MW/month** or **₹/kWh** basis
- Approved annually by the Telangana Electricity Regulatory Commission (TGERC)

Wheeling Charges

Wheeling charges are levied by the DISCOM for using the **distribution network** (usually 33 kV and below) to deliver power to end consumers.

Wheeling charges:

- Cover operation and maintenance of distribution infrastructure
- Are typically higher at lower voltage levels
- Are applied per unit of energy wheeled (₹/kWh)

Key Point

Even if the generator is connected at a high voltage level, power is ultimately delivered to consumers through the **distribution network**, so wheeling charges generally apply for final delivery.

10.2 Transmission and Wheeling Losses

Losses represent the **physical loss of electrical energy** that occurs when power flows through transmission and distribution systems.

Transmission Losses

- Occur on high-voltage lines and substations
- Caused by conductor resistance, transformer losses, and reactive power flow
- Applied as a **percentage deduction** from energy injected

Wheeling (Distribution) Losses

- Occur on 33 kV, 11 kV, and LT networks
- Higher at lower voltage levels due to higher current and longer feeder lengths

Typical Loss Ranges (Indicative, Telangana)

- Transmission losses: ~2–4%
- Wheeling losses (33 kV): ~3–6%
- Wheeling losses (11 kV / LT): higher and consumer-specific

Loss percentages are **not negotiated**, but notified annually by the regulator and applied uniformly.

Important Clarification

Losses are **not charges** paid in money.

They are applied as a **reduction in energy units** available for billing and settlement.

10.3 Cross Subsidy Surcharge (CSS) and Additional Surcharge (AS)

Cross Subsidy Surcharge (CSS)

CSS is a regulatory charge levied to compensate DISCOMs when high-paying industrial consumers procure power from sources other than the DISCOM.

CSS applies to:

- **Third-party Open Access consumers**

CSS does **not apply** to:

- Captive Open Access (when compliant)
- Group Captive Open Access (when compliant)

CSS is notified:

- Consumer category-wise
- Voltage-wise
- Annually through tariff orders

Additional Surcharge (AS)

AS is intended to recover fixed costs incurred by DISCOMs due to long-term power procurement commitments.

As per **recent TGERC orders (2025)**:

- Additional Surcharge for Open Access consumers has been **notified as NIL** for the applicable period

However:

- AS is subject to periodic review
- It may be reintroduced in future tariff periods

Why Captive and Group Captive Matter Commercially

- Captive and Group Captive projects **avoid CSS and AS**
- This exemption is one of the **largest contributors to cost savings**
- Non-compliance with captive conditions can lead to **retrospective levy**

10.4 Banking Charges and Lapse Rules

Banking allows renewable energy generated at one time to be **adjusted against consumption at another time**, subject to regulatory limits.

In Telangana:

- Banking is generally permitted on a **monthly basis**
- A portion of banked energy is deducted as **banking charges (in kind)**
- Unutilized banked energy **lapses** at the end of the banking cycle

Banking is:

- A **commercial adjustment**, not physical storage
- Intended to help manage mismatch between solar generation hours and industrial load profiles

Commercial Implication

- Banking improves flexibility
- Excessive reliance on banking can reduce effective savings due to:
 - Banking charge deductions
 - Lapse of unused energy

10.5 Illustrative Billing Build-Up (Conceptual)

A simplified billing flow for an Open Access consumer:

1. Gross energy generated at the plant (kWh)
2. Deduct transmission losses
3. Deduct wheeling losses
4. Apply banking adjustments (if applicable)
5. Allocate net energy to consumer(s)
6. Apply applicable charges:
 - Transmission charges
 - Wheeling charges
 - CSS (if applicable)
 - AS (if applicable)
7. Net payable amount determined

The **difference between DISCOM tariff and net landed Open Access cost** represents the consumer's savings.

11. Curtailment and Grid Constraints

Curtailment refers to the **reduction of power generation from a solar plant**, even when sunlight is available, due to **grid-related constraints**.

In Open Access projects, curtailment is a **key commercial and operational risk** that must be clearly understood and planned for.

11.1 Causes of Curtailment

Curtailment in Telangana primarily occurs due to **technical and operational limitations of the grid**, not due to lack of demand from consumers.

The most common causes include:

Grid Congestion

- Occurs when multiple generators inject power simultaneously into the same transmission or distribution corridor
- Common during **midday hours**, when solar generation peaks across multiple plants
- Lines, transformers, or substations reach their safe loading limits

Substation Capacity Constraints

- Limited transformer capacity
- Insufficient available bays
- Thermal limits of equipment at the substation

Voltage Rise Issues

- High solar injection at lower voltage levels (11 kV / 33 kV) can cause voltage to exceed permissible limits
- To protect consumer equipment and grid stability, generation is backed down

Transmission Corridor Limitations

- Power must flow through predefined corridors
- If downstream evacuation capacity is not sufficient, upstream plants are curtailed

System Security and Stability

- Grid operator may curtail generation to maintain:
 - Frequency stability

- Voltage stability
- Safe operating margins

Curtailment is **not linked to the consumer's willingness to consume power**, but to the grid's ability to carry and absorb it at that moment.

11.2 Operational Handling by SLDC

Curtailment decisions are made and implemented by the **State Load Dispatch Centre (SLDC)**.

Key aspects of SLDC handling include:

- SLDC monitors real-time grid conditions across:
 - Generation
 - Load
 - Line flows
 - Voltage levels
- When constraints are identified, SLDC issues:
 - Back-down instructions
 - Generation limits (MW caps)
 - Time-specific curtailment orders

For solar and wind:

- These resources are generally treated as “**must-run**”
- Must-run means:
 - They are **not curtailed for economic reasons**
 - They **can still be curtailed for grid safety or congestion reasons**

Curtailment instructions are:

- Time-bound
- Location-specific
- Mandatory for compliance

Non-compliance can lead to penalties or disconnection.

11.3 Commercial Impact on Projects

Curtailment directly impacts the **financial performance** of Open Access projects.

Key impacts include:

Reduced Energy Delivery

- Units that could have been generated are not produced
- Leads to lower billed energy for consumers

Lower Effective CUF

- Curtailment reduces annual generation
- This lowers realized CUF compared to design CUF

Impact on Consumer Savings

- Less solar energy available to offset grid power
- Consumers may need to draw more power from DISCOM at higher tariffs

Mismatch with PPA Expectations

- PPAs are usually signed assuming a certain annual energy
- Curtailment introduces uncertainty unless addressed contractually

Banking and Settlement Effects

- Reduced generation limits the ability to:
 - Bank energy
 - Optimize time-of-day usage

Curtailment risk is **location-specific** and can vary significantly between substations and voltage levels.

11.4 Curtailment Mitigation Strategies

Curtailment cannot be fully eliminated, but it can be **significantly reduced through planning and structuring**.

Location and Substation Selection

- Prefer substations with:
 - Available capacity
 - Lower existing solar penetration
- Avoid heavily saturated feeders and substations

Higher Voltage Connectivity

- Connecting at **132 kV or above** generally:
 - Reduces congestion risk
 - Provides access to stronger transmission corridors
- Though costlier, it improves long-term reliability

Conservative Generation Assumptions

- Financial models should assume:
 - Some level of curtailment
 - Lower effective CUF than theoretical values

PPA Structuring

- Clearly define:
 - Curtailment treatment
 - Whether deemed generation applies (if at all)
 - How shortfall is handled

Diversified Consumer Base

- Group captive structures with multiple consumers:
 - Improve absorption of available energy
 - Reduce dependence on a single load profile

Hybrid and Storage Roadmap

- While still emerging, future mitigation options include:
 - Battery storage
 - Hybrid solar-wind configurations
- These help shift energy away from congested hours

Curtailment should be treated as a **core project risk**, not an exception.

12. Contracting and Compliance

This section explains the **contractual framework and compliance requirements** governing solar Open Access projects.

While technical execution enables power flow, **contracts and compliance ensure that power delivery, payments, and regulatory benefits remain legally valid over the project life.**

12.1 Power Purchase Agreement (PPA) Structure

The Power Purchase Agreement (PPA) is the primary commercial contract between the **solar power generator (or SPV)** and the **power consumer(s)**.

For Open Access projects, the PPA typically defines:

- Contracted capacity (MW)
- Energy supply obligations
- Tariff structure
- Billing and payment terms
- Risk allocation between parties

Key clauses commonly included in Open Access PPAs:

- **Tariff and escalation**
 - Fixed tariff or indexed tariff over the contract term
- **Contract tenure**
 - Typically 10–25 years for solar projects
- **Scheduling and delivery**
 - Reference to SLDC schedules and metered energy
- **Curtailement treatment**
 - Whether curtailed energy is deemed delivered or not
- **Banking and lapse**
 - Treatment of banked or unutilized energy
- **Force majeure**
 - Natural events, grid failures, regulatory actions
- **Change in law**
 - Allocation of impact due to regulatory changes
- **Payment security**
 - Letter of Credit, escrow, or other mechanisms
- **Termination and default**
 - Conditions under which the agreement can be terminated

For **third-party Open Access**, the PPA is the sole commercial contract.

For **captive and group captive**, the PPA works in conjunction with ownership and compliance agreements.

12.2 Group Captive Shareholder Agreements

In **group captive projects**, consumers are also **equity participants** in the generating company (SPV).

This requires a **Shareholder Agreement (SHA)** in addition to the PPA.

The SHA governs:

- Equity ownership structure
- Voting rights
- Capital contribution obligations
- Transfer restrictions
- Exit mechanisms

Key elements relevant for compliance:

- **Minimum 26% equity ownership**
 - Must be collectively held by captive users
 - Must include voting rights
- **Energy consumption linkage**
 - Shareholding must broadly align with energy offtake
- **Lock-in provisions**
 - Prevent dilution or transfer that could break captive status
- **Default handling**
 - What happens if a member fails to consume allocated power
- **New member induction**
 - Process to add or replace captive users

The SHA ensures that **captive qualification is maintained not just commercially, but legally**.

12.3 Captive Generating Plant (CGP) Compliance and Verification

To qualify as a **Captive Generating Plant (CGP)** under Indian electricity regulations, **both ownership and consumption conditions must be met**.

Key CGP conditions:

- **Ownership condition**
 - Captive users must own at least **26% of equity with voting rights**
- **Consumption condition**
 - Captive users must consume at least **51% of total annual generation**

Compliance characteristics:

- Assessed on an **annual basis**
- Based on:
 - Shareholding records
 - Metered energy consumption
- Verified by utilities and auditors as required

Consequences of non-compliance:

- Loss of captive status
- Retroactive application of:
 - Cross Subsidy Surcharge (CSS)
 - Additional Surcharge (AS)
- Potential contractual penalties under PPA/SHA

Because of this, **energy allocation, scheduling, and drawal discipline are critical** for captive and group captive projects.

12.4 Change-in-Law and Regulatory Risk

Electricity regulations are subject to periodic change through:

- Tariff orders
- Policy amendments
- Judicial rulings
- Regulatory clarifications

Examples of change-in-law impacts:

- Introduction or removal of banking
- Changes in CSS or AS rates
- Modification of loss factors
- Curtailment rules
- Eligibility conditions for captive status

PPAs typically include a **change-in-law clause** that defines:

- What constitutes a change in law
- How additional costs or benefits are shared

- Whether tariffs are adjusted or compensated
- Notification and claim timelines

This clause is critical for **long-term project bankability**, especially for 15–25 year contracts.

12.5 Payment Security

Payment security is one of the **most critical commercial risks** in Open Access projects, because:

- Power is delivered continuously
- Payments are typically made **after consumption**
- There is no DISCOM-backed payment guarantee for OA power

12.5.1 What is Payment Security?

Payment security refers to mechanisms that ensure the generator receives payment for electricity supplied, even if the consumer delays or defaults.

Unlike DISCOM PPAs, Open Access projects rely entirely on **private contractual enforcement**.

12.5.2 Common Payment Security Mechanisms

Typical mechanisms used in Open Access PPAs include:

- **Security Deposit**
 - Cash deposit equal to 1–3 months of expected billing
- **Letter of Credit (LC)**
 - Issued by a bank in favor of the generator
 - Covers a defined billing amount
- **Bank Guarantee (BG)**
 - Callable on payment default
- **Escrow Account**
 - Consumer deposits revenue into a controlled account
- **Advance Payment**
 - Partial or full advance payment model (less common)

The strength of payment security often depends on the **credit profile of the consumer**.

12.5.3 Credit Risk in Open Access Projects

Credit risk is the risk that a consumer:

- Delays payment

- Disputes bills
- Becomes financially stressed
- Shuts down operations or relocates

Credit risk is higher when:

- Consumers are small or financially weak
- Group captive projects have many members
- Long payment cycles are allowed
- There is no strong termination protection

Unlike DISCOM supply, Open Access does **not provide sovereign or regulatory payment backing**.

12.5.4 Payment Risk in Group Captive Projects

In group captive projects:

- Multiple consumers share generation
- Default by one consumer can affect project cash flows
- Equity ownership does not guarantee timely payment

Mitigation approaches:

- Individual payment security from each captive user
- Right to reallocate power to alternate consumers
- Step-in rights for the generator
- Clear default and replacement clauses in PPA and SHA

12.5.5 Consequences of Payment Default

If payment security is weak, consequences may include:

- Cash flow stress for the project
- Inability to service debt
- Operational disruptions
- Legal disputes and arbitration
- Loss of lender confidence

Well-structured payment security is therefore **essential for project bankability**.

13. End-to-End Case Studies

This section illustrates how solar Open Access projects work in practice, from contracting to billing,

using two realistic Telangana-based examples.

13.1 Case Study 1 – 5 MW Solar Plant at 33 kV Serving Multiple Industries

Project Overview

- Plant capacity: **5 MWAC**
- DC capacity: **6.25 MWp** (DC/AC ratio = 1.25)
- Grid connection voltage: **33 kV**
- DISCOM network
- Business model: **Group Captive Open Access**
- Location of plant: Northern Telangana
- Consumers: Multiple industries in different locations under same DISCOM

Step 1: Consumer Contracts and Structure

- 5 industrial consumers (Factories A–E)
- Each consumer:
 - Holds equity in the SPV
 - Collective equity $\geq 26\%$
 - Collective annual consumption $\geq 51\%$
- PPAs signed with each consumer for:
 - Energy charge (₹/kWh)
 - Banking terms
 - Curtailment sharing
 - Settlement timelines

Step 2: Solar Plant Setup (Generation Side)

Solar Modules

- Total DC capacity: **6.25 MWp**
- Example: ~11,400 modules $\times$ 550 Wp

Inverters

- Inverter capacity: **5 MWAC**

- Example: 10 string inverters × 500 kVA each
- Power factor setpoint: **0.98 lag/lead**
- Reactive power support enabled

Why DC > AC

- Higher energy generation during morning/evening
- Better CUF
- Some clipping during peak noon hours accepted

Step 3: Transformers and Internal Electrical System

Inverter Transformers

- Step up from **800 V AC** → **33 kV**
- Typical rating: 5.5–6 MVA

Main Electrical Components

- Internal HT switchgear (33 kV)
- Protection relays
- Metering panel (ABT/SEM compliant)

Step 4: Evacuation and Grid Interconnection

Evacuation Line

- 33 kV overhead line from plant to nearest DISCOM substation
- Length example: 5–10 km
- Built by developer

Substation Interface

- At 33 kV:
 - Bay may or may not be required depending on substation design
 - Usually a **dedicated feeder with protection and metering**
- DISCOM approval required for:
 - Short-circuit levels
 - Voltage regulation

- Protection coordination

Step 5: Energy Generation and Expected Output

Generation Assumptions

- CUF: **20%**
- Annual generation:
 - $5 \text{ MW} \times 8,760 \text{ hrs} \times 20\%$
 - $\approx \text{8.76 million kWh (8.76 MU)}$

Step 6: Scheduling and Injection

- Day-ahead generation schedule submitted to SLDC
- Solar treated as **must-run**
- Actual injection measured at plant meter (15-minute blocks)

Step 7: Drawal at Consumer End

- Each factory draws power from its local DISCOM substation
- No physical tracing of electrons
- Drawal measured at consumer meters

Step 8: Losses and Banking

- Transmission loss (33 kV): $\sim 1\text{--}2\%$
- Wheeling loss: $\sim 3\text{--}4\%$
- Net deliverable energy $\approx \text{94--96\%}$
- Excess midday generation:
 - Banked (subject to Telangana banking rules)
 - Adjusted against evening/night consumption

Step 9: Billing and Settlement

- Monthly settlement by DISCOM/SLDC:
 - Injection energy
 - Loss adjustments

- Allocation to each consumer
- Banking adjustments
- Consumers billed based on:
 - Net allocated energy
 - PPA tariff
- CSS and AS: **Not applicable** (group captive compliant)

Step 10: Key Risks

- Curtailment during high solar hours
- Banking restrictions
- Captive compliance shortfall if consumption drops
- Substation congestion

13.2 Case Study 2 – 10 MW Solar Plant at 132 kV Serving Industries Across DISCOMs

Project Overview

- Plant capacity: **10 MWAC**
- DC capacity: **13 MWp** (DC/AC ratio = 1.3)
- Grid voltage: **132 kV**
- TRANSCO network
- Business model: **Third-Party Open Access**
- Plant location: Solar-rich northern Telangana
- Consumers: Large industries across different districts

Step 1: Consumer Contracts

- PPAs signed with multiple large consumers
- No equity ownership by consumers
- Energy price lower than DISCOM tariff
- CSS and AS applicable

Step 2: Solar Plant Setup

Solar Modules

- DC capacity: **13 MWp**
- High-efficiency modules (540–600 Wp)

Inverters

- Central or large string inverters
- Total inverter capacity: **10 MWAC**
- Inverter PF controlled dynamically by SLDC instructions

Step 3: Transformers and Switchyard

Inverter Transformers

- Step up: 800 V → 33 kV

Main Step-Up Transformer

- Step up: 33 kV → 132 kV
- Rating: ~12.5–15 MVA

Switchyard (132 kV)

- Mandatory bay at TRANSCO substation
- Includes:
 - Breakers
 - Isolators
 - CTs / PTs
 - Protection panels
 - Revenue metering

Step 4: Evacuation

- 132 kV transmission line to nearest TRANSCO substation
- Right-of-Way permissions required
- Higher cost but:
 - Lower losses
 - Higher power transfer capability
 - Lower curtailment risk

Step 5: Generation and Output

- CUF: **21%**
- Annual energy:
 - $10 \text{ MW} \times 8,760 \times 21\%$
 - **$\approx 18.4 \text{ MU}$**

Step 6: Scheduling and Injection

- SLDC scheduling mandatory
- Curtailment possible during congestion
- Higher priority corridor compared to 33 kV

Step 7: Drawal Across Locations

- Consumers draw power from their local DISCOM substations
- Power flows across DISCOM boundaries
- Settled commercially via SLDC accounting

Step 8: Charges and Billing

- Applicable charges:
 - Transmission charges
 - Wheeling charges
 - CSS
 - AS (if applicable)
- No captive exemptions
- Monthly settlement statements issued

Step 9: Key Risks

- Higher upfront cost
- Regulatory change risk (CSS/AS)
- Long-term PPA credit risk
- Transmission corridor congestion

13.3 Key Takeaways from Both Case Studies

- 33 kV projects:
 - Lower capex
 - Higher congestion risk
 - Suitable for ≤ 5 –7 MW
- 132 kV projects:
 - Higher capex
 - Lower losses
 - Better scalability
- Commercial settlement is **always accounting-based**
- Physical power flow follows grid physics, not contracts
- Proper planning of:
 - Voltage level
 - Substation
 - Consumer mix

is critical for long-term success

14. Checklists

This section provides **practical checklists** and **reference sources** to support planning, execution, and compliance of solar Open Access projects in Telangana.

These checklists are intended to reduce execution risk and ensure regulatory readiness.

14.1 Site and Substation Selection Checklist

Land and Site Suitability

- Adequate land area based on plant capacity (approx. 4–5 acres per MW for fixed tilt)
- Clear land title and right to use for project life
- Terrain suitable for construction (minimal leveling required)
- Flood risk and soil bearing capacity assessed
- Proximity to existing substations or feeders

Grid Proximity and Voltage

- Nearest substation identified with voltage level (33 kV / 132 kV / 220 kV)
- Available capacity at substation confirmed (not just theoretical capacity)
- Distance to substation assessed for evacuation line cost and losses
- Voltage level selected considering:
 - Plant size
 - Curtailment risk
 - Capital cost

Evacuation Feasibility

- Right of Way (RoW) feasibility for evacuation line
- Crossing requirements (roads, canals, railways, private land)
- Expected timeline for line construction
- Requirement of bay confirmed based on voltage level

14.2 Open Access Approval Checklist

Pre-Development Approvals

- Grid connectivity application submitted
- Feasibility / connectivity approval received
- Substation identified and approved
- Voltage level and injection point finalized

Construction & Commissioning

- Evacuation line approval
- Bay approval (if applicable)
- Metering scheme approval
- Protection scheme approval
- SCADA / telemetry approval (if required)

Operational Approvals

- Open Access approval from DISCOM
- Scheduling approval from SLDC

- Energy accounting and settlement approval
- Final synchronization and commercial operation declaration (COD)

14.3 Group Captive Compliance Checklist

Ownership Compliance

- Minimum 26% equity with voting rights held by captive users (collectively)
- Shareholding structure documented and traceable
- No dilution below threshold during the year

Consumption Compliance

- Captive users collectively consume at least 51% of annual generation
- Individual allocation ratios documented
- Monthly tracking of generation vs consumption

Documentation and Reporting

- Shareholding certificates and registers
- Energy generation and drawal statements
- Annual captive compliance statement
- Readiness for regulatory or audit verification

Risk Controls

- Mechanism to reallocate power if one consumer under-draws
- Backup consumers identified where possible
- Contractual provisions for non-compliance scenarios

Reference Documents and Sources

Regulatory and government sources

- [TGERC Official Website \(Regulations & Orders\)](#)

- [Terms and Conditions of Open Access Regulation 2024 \(TGERC Regulation 1 of 2024\)](#)
- [Open Access Regulation Gazette Notification](#)
- [Wheeling Tariff Order FY 2025-26 \(wheeling charges/losses\)](#)
- [Transmission Tariff Order FY 2025-26 \(transmission charges\)](#)
- [Additional Surcharge Order H2 FY 2025-26 \(AS = nil for Open Access\)](#)
- [List of Tariff Orders & filings for FY 2025-26](#)

Policy & tertiary sources

- [Telangana Clean and Green Energy Policy 2025](#)
- [Mercom India – Telangana Rooftop & Open Access commentary](#)
- [Green Energy Open Access Policy Overview — Novergy Solar](#)